

BahriApp — software record

Authoritative provenance document for the BahriApp software archive. Covers both archived versions: what each is, what each produced, how they differ, and which one any given claim or dataset belongs to.

Included in both version archives so that either one, downloaded alone, tells the whole story.

Software	BahriApp — multimodal behavioural-biometric acquisition platform
Archived versions	v1.0.0, v1.1.0
Licence	MIT (both)
Concept DOI	10.5281/zenodo.XXXXXXX
Article	SoftwareX, 10.1016/j.softx.XXXX.XXXXX
Dataset	10.5281/zenodo.XXXXXXX
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1. Which version do I want?

If you are...	Use	DOI
Collecting new data	v1.1.0	version DOI
Analysing the pilot corpus	v1.0.0 — the code that produced it	version DOI
Reproducing the article's measurements	v1.0.0 for corpus figures, v1.1.0 for the correction figures	both
Citing the project generally	concept DOI	resolves to latest
Reviewing the audit	v1.1.0 → <code>analysis/</code> , <code>CHANGELOG.md</code>	version DOI

The most common mistake to avoid: taking v1.1.0 and assuming it produced the pilot corpus. It did not. Eighteen exported columns mean something different — or nothing — in data collected with v1.0.0.

2. Timeline

2025-01-01	Pilot data collection, v1.0.0 454 enrolled · 265 contributing · 341,606 records
2025-05-07	
2026	— SoftwareX submission S0FTX-D-26-00834 — Peer review: two algorithmic defects identified — Audit of v1.0.0 against the corpus it produced — 13 further defects found, disclosed
2026-09-19	— v1.1.0 released; both versions archived

3. What is common to both versions

Unchanged between v1.0.0 and v1.1.0:

- **Eleven modalities** in one application: fixed-text, free-text and password keystroke dynamics in English and Amharic; swipe; tap; handwriting trajectories in two scripts; accelerometer gait; gyroscopic tilt.
- **Native Amharic keystroke capture.** The system keyboard is suppressed and the application renders its own two-tier fidel keyboard, so every logged event is a physical touch on a key the app drew. No transliteration IME, no candidate list, no OS composition. A first-order fidel costs one key event; any other order costs two.
- **Gamified elicitation** across six game categories, with individual and team leaderboards.
- **Offline-first synchronisation** via an on-device Hive queue.
- **Bahri Panel** desktop dashboard with CSV export.
- **Flutter/Dart** client targeting Android 8.0 (API 26) and above.
- **MIT licence.**

4. Version 1.0.0

4.1 Role

The version that collected the pilot corpus. Archived **unmodified**; see `ARCHIVAL_NOTICE.md` in that archive for why, and for the integrity manifest covering all 469 original files.

4.2 What it produced

Acquisition window	1 January – 7 May 2025 (125 days)
Enrolled participants	454
Contributing participants	265 (58.4 %)
Records	341,606
Sessions	23,106
Keystroke session vectors	4,721
Exported feature columns	165
Sites	Three Ethiopian higher-education institutions, BYOD

Per-modality yield is in the article's Table 3 and in `analysis/ §3`.

4.3 Known defects

Fifteen classes, found by auditing this code against its own output. Two were identified by reviewers; the rest by the audit their comments prompted.

#	Component	Defect	Data consequence
1	Gyroscope filter	Scalar <code>20</code> compared against two angular velocities (rad/s) and one angular acceleration (rad/s ²)	Retained 98.53 % of samples, 98.41 % by the acceleration term alone. Stream is effectively unfiltered.
2	Gyroscope orientation	Gravity-vector tilt formulas applied to angular velocity	<code>roll / pitch</code> are not orientation. Valid only as rotation-axis direction.
3	Δt computation	<code>Duration.inMilliseconds</code> truncates; <code>> 0</code> guard then drops faster samples	Sub-millisecond events silently discarded, uncounted.
4	Clock	Wall-clock <code>DateTime.now()</code>	Hold times to -1040 ms; negative flight times.
5	Swipe collector	<code>updateCollecting()</code> an empty stub	No trajectory retained; path features uncomputable.
6	Swipe features	Fling velocity (px/s) assigned to <code>endX / endY</code> , used as coordinates (px)	9 columns mix units. Median "distance" = 2.8 screen diagonals.
7	Swipe features	<code>finger0orientation</code> duplicates <code>angle</code> ; <code>deceleration</code> duplicates <code>acceleration</code> ; time-of-day impact constant	$r = 1.000000$; one column has a single value.
8	Tap rate	<code>1 ~/ duration</code> — integer division	Exactly 0 in 99.29 % of records.
9	Tap drift	x term repeated where y intended	Median 421 px, floor 102 px.
10	Tap latency	Release time passed as screen-response time	Duplicates duration, $r = 0.999997$.
11	Keystroke rates	Division by <code>Duration.inSeconds</code>	Sub-second sessions divide by zero.
12	Keystroke aggregation	No pause segmentation; unguarded CV; three aliased metrics	37-min gap as one flight time; negative CVs; pairwise $r \geq 0.9975$.
13	Gait orientation	Magnitude series indexed as three axis components	One constant orientation per session, all 489 sessions.
14	Gait jerk / cap	Jerk sampled only at downward zero-crossings; 20-step cap	100 % negative; 43.4 % of sessions censored.
15	Tap export	Epoch columns written in scientific notation	4,150 rounds $\rightarrow$ 537 apparent sessions.

4.4 Retracted columns

Eighteen of 165. Four recomputable, fourteen not.

Modality	Count	Recomputable?
Swipe	10	No — trajectory never stored
Tap	2	Yes — from stored coordinates and duration
Gait	4	No — raw axes never stored
Gyroscope	2	Reinterpretable, not orientation

Full list with reasons and corrected definitions: `docs/data_dictionary.md` in v1.1.0.

4.5 Security posture

No encryption of any kind was implemented. The v1.0 README's claim of end-to-end encryption is withdrawn in full.

Aspect	v1.0.0
Transport	Plain HTTP to a hard-coded IPv4 address, 24 endpoints
On-device cache	Eight Hive boxes, no <code>encryptionCipher</code>
Authentication	Token stored at login, never attached to uploads

Key custody	None
Export	Included an <code>Email</code> column
Credentials	Absolute developer path

Implication for the corpus: data travelled unencrypted and sat unencrypted on devices until upload. Because uploads were unauthenticated, records cannot be cryptographically attributed. No evidence of interception has been found; that is not evidence of absence.

Participants consented to behavioural collection, minimal demographics, and publication of a de-identified corpus, and consent materials stated that behavioural biometrics can act as persistent identifiers.

5. Version 1.1.0

5.1 Role

Current. Use for new data collection. Corrects all fifteen defect classes.

5.2 Corrections

Area	v1.0.0	v1.1.0
Gyroscope gate	One scalar, three quantities, two dimensions	Per-dimension thresholds; <code>calibrate()</code> derives the speed gate from observed data (pilot $p70 = 1.63 \text{ rad/s} = 93 \text{ }^\circ/\text{s}$)
Orientation	<code>atan2</code> of angular velocity	Complementary filter fusing gyroscope + accelerometer; honest <code>omegaDirection()</code> retained separately
Clock	Wall clock, ms	Monotonic, μs
Δt	Truncated, unguarded	Explicit window [0.1 ms, 250 ms]; rejects and counts
Swipe	Two points	Full trajectory; dimensionally consistent kinematics
Tap	Integer division, axis typo	Float rate; <code>.dy</code> fixed; drift and targeting error separated
Keystroke	11 scalars, unguarded	28 statistics, pause segmentation at 3 s, guarded CV
Gait	Magnitudes as axes; signed jerk; 20-step cap	Three-axis orientation; signed jerk over the full cycle; cap removed
Export	Floats	Quoted strings + post-write verification
Instrument metadata	None	Device model, screen geometry, realised rate, jitter, rejection counts

5.3 Security

Control	Implementation
Transport	HTTPS enforced at startup; optional SHA-256 certificate pinning
Cache	<code>HiveAesCipher</code> , 256-bit key in the Android Keystore, purged on ack
Authentication	Bearer token on every request; refresh rotation
Replay	<code>Idempotency-Key</code> + server-side unique constraint
Export	Identifier columns removed; role-gated; append-only audit log
Withdrawal	<code>withdraw_participant()</code> cascades across behavioural tables

An eight-threat model is in `SECURITY.md`. **De-identification is partial, not absolute** — behavioural biometrics are themselves persistent identifiers.

5.4 Verification shipped with the release

Artefact	What it checks
<code>test/feature_extractors_v11_test.dart</code>	30 unit tests, 5 groups. Each encodes a property v1.0 violated, so the suite fails against v1.0 by construction.
<code>analysis/verify_extractor_logic.py</code>	Same 30 assertions in Python — runs without a Flutter toolchain.
<code>tools/audit_v10_defects.sh</code>	15 defect-class sweep; all report CLEAR.
<code>analysis/...ipynb</code>	Regenerates 31 result tables and 8 figures from the released corpus.

5.5 Known limitations

- Android only.
- The v1.0 corpus cannot be retrospectively repaired for the fourteen non-recomputable columns.
- The pilot had no non-gamified control arm, so its retention figures cannot establish that gamification reduced attrition.
- v1.1.0's Dart sources were not compiled at package-build time (no toolchain available); the arithmetic was verified in Python

instead. Run `flutter analyze` and `flutter test` before deployment.

- `pubspec.lock` is not in the archive — the v1.0 `.gitignore` had a blanket `*.lock` rule. Generate with `flutter pub get` and commit.

6. Relationship to the publications

Output	Describes	Cite
SoftwareX article	The tool , at v1.1.0, with pilot figures from v1.0.0 data	article DOI
Companion dataset publication	The corpus , collected with v1.0.0	dataset DOI
This software record	Both versions	concept DOI

The 578-feature registry referred to in the initial submission is a downstream feature-engineering product belonging to the dataset publication, not to the acquisition tool. The tool exports 165 columns. The article's §2.2 reconciles the counts.

7. Reproducing the article's numbers

1. Obtain the pilot corpus (dataset DOI).
2. Place the eleven CSVs in `analysis/data/`.
3. Run the notebook from **v1.1.0** — it audits v1.0.0 output and needs the corrected reference implementations to compute the "after" figures.
4. Expect: `RAN 30 | failures 0 | errors 0` from the logic verifier, and `headline_numbers.json` matching the article.

Key figures for a quick check:

Quantity	Value
Contributing / enrolled	265 / 454 = 58.4 %
Acquisition span	125 days
Records	341,606
Gyro records retained by the v1.0 filter	98.53 %
...by the acceleration term alone	98.41 %
Records exceeding the 20 rad/s gate	0.117 %
Cross-modal session overlap	0.61 %
Amharic / English key events per rendered char	1.44 / 0.94
Recovered device profiles	71, spanning 611–1468 logical px

8. Archive contents

v1.0.0

```
BahriApp-v1.0.0/  
├─ ARCHIVAL_NOTICE.md      ← read first  
├─ MANIFEST.sha256         ← 469 original files + 4 additions  
├─ CITATION.cff  
├─ .zenodo.json  
├─ docs/SOFTWARE_RECORD.md  
├─ tools/fill_dois.py  
└─ [469 original source files, unmodified]
```

v1.1.0

```
BahriApp-v1.1.0/  
├─ README.md · CHANGELOG.md · MIGRATION.md · SECURITY.md · DEPLOY.md  
├─ CITATION.cff · .zenodo.json · .env.example  
├─ lib/  
│ └─ config/app_config.dart  
│   └─ services/v11/      ← corrected extractors + security layer  
├─ test/feature_extractors_v11_test.dart ← 30 tests  
├─ docs/  data_dictionary · comparison_protocol · ZENODO_ARCHIVE · SOFTWARE_RECORD  
├─ analysis/  notebook · figures · logic verifier  
├─ deploy/    docker-compose · schema.sql · nginx.conf  
├─ example_data/  synthetic CSVs + generator  
└─ tools/      audit_v10_defects.sh · fill_dois.py
```

9. Citing

Cite the **version you used**.

```

@software{aseres_bahriapp_v1_1_0,
  author = {Aseres, Animaw Kerie and Beyene, Asrat Mulatu
    and Tegegne, Lemlem Kassa},
  title  = {BahriApp: An Android Platform for Gamified Multimodal
    Behavioural-Biometric Data Acquisition, v1.1.0},
  version = {1.1.0}, year = {2026},
  doi     = {10.5281/zenodo.XXXXXXX}
}

@software{aseres_bahriapp_v1_0_0,
  author = {Aseres, Animaw Kerie and Beyene, Asrat Mulatu
    and Tegegne, Lemlem Kassa},
  title  = {BahriApp: An Android Platform for Gamified Multimodal
    Behavioural-Biometric Data Acquisition, v1.0.0},
  version = {1.0.0}, year = {2025},
  note    = {Version used to collect the pilot corpus},
  doi     = {10.5281/zenodo.XXXXXXX}
}

```

Machine-readable equivalents are in each archive's CITATION.cff .